Practice workbook · Getting terms out of a formula for the sum

Getting the first term, common difference or common ratio, or a particular sum, directly from a formula for S_n or a sigma expression.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-02

1. Question 6 · part a · 4 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part a only.

6 An arithmetic series A has first term a and common difference d.

The sum S_n of the first n terms of A is given by $S_n = n(15 + 2n)$

(a) Find the value of a and the value of d.

(4)

(b) Find the 20th term of A.

(2)

Given that $S_{2p} - 2S_p = 1 + S_{(p-1)}$

(c) find the value of p.

(4)

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1. Question 6 · part a · 4 marks · Working and answer

Working and answer

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2. Question 6 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

6 The sum of the first n terms of an arithmetic series is S_n where $S_n = n(3 + 2n)$

(a) Find the value of S_{20}

(2)

Given that $S_n = \sum_{r=1}^{n} (Ar + B)$

(b) find the value of A and the value of B

(6)

A different arithmetic series has first term 7 and common difference 4

The sum of the first n terms of this series is T_n

(c) Use algebra to find the value of n for which $T_n = S_n + 252$

(5)

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2. Question 6 · part a · 2 marks · Working and answer

Working and answer

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3. Question 7 · part a.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part a.i only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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3. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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4. Question 7 · part a.ii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part a.ii only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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4. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 7 · part b.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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5. Question 7 · part b.i · 0 marks · Working and answer

Working and answer

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6. Question 7 · part b.ii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference. **(2)**

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio. **(2)**

(c) Find the value of n for which $A_{14} - 6 = G_n$ **(5)**

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6. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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